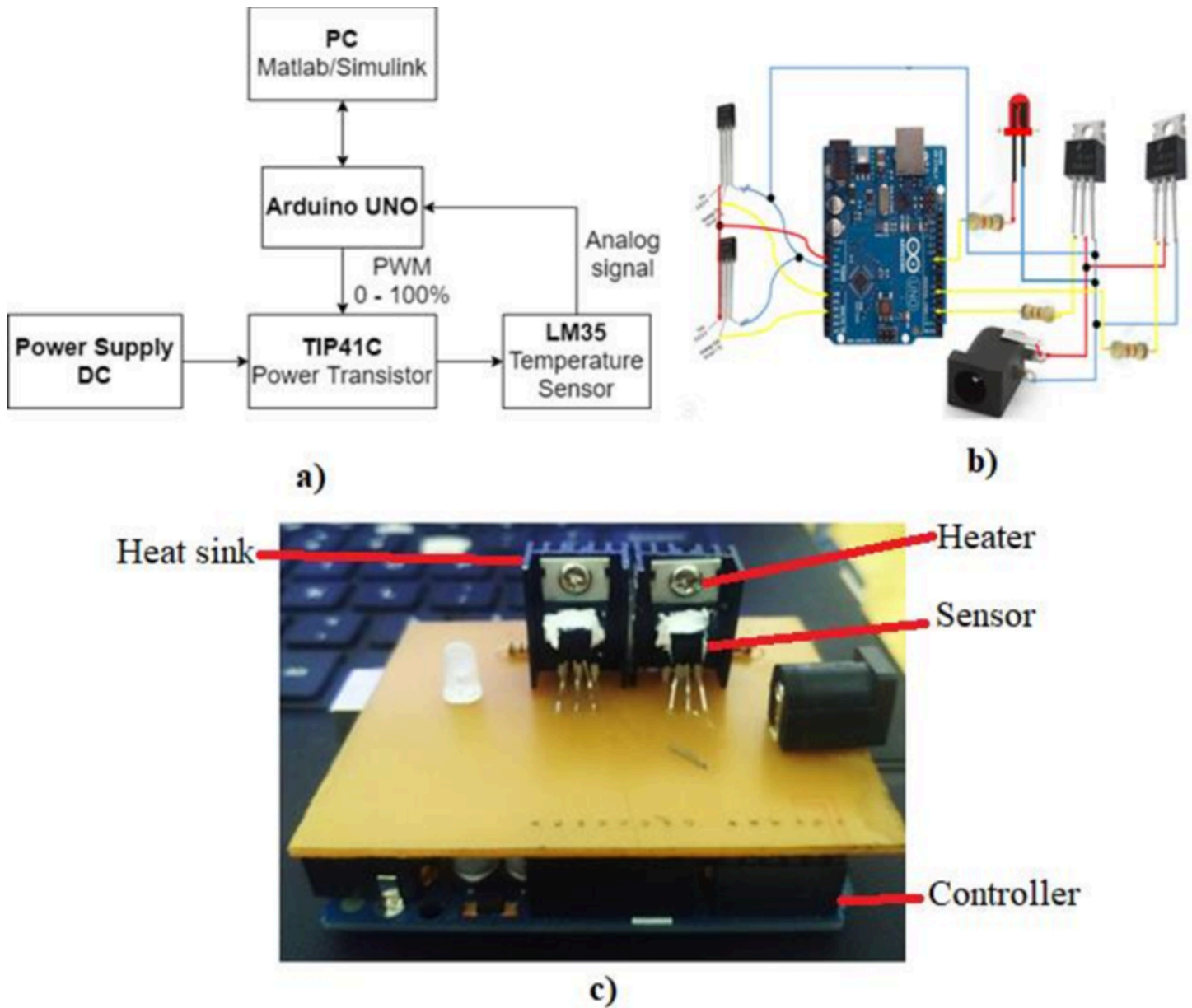


Control Systems and Applications

Defense of the last milestone

This will be the last of the Milestones used to evaluate this course.



For the last milestone a TCLab kit will be assigned to each group, together with a portable DC power supply and a usb cable. Each TCLab kit consists of an arduino and a powerboard with two transistors TIP41C with heatsinks and two temperature sensors LM35.

An extensive amount of software tools, code and inspiration can be found at:

<http://apmonitor.com/pdc/index.php/Main/ArduinoTemperatureControl>

Procedure

For these steps you can use python or matlab. The choice is entirely yours.

System Identification of the two subsystems

TCLab has the following key components:

Heater 1 (H1) and **Heater 2 (H2)**: two independent heaters.

Temperature Sensor 1 (T1) and **Temperature Sensor 2 (T2)**: two temperature sensors measuring the temperatures close to each respective heater.

Lets define:

$$G_{\{1\}}(s) = \frac{T_{\{1\}}}{H_{\{1\}}}$$

$$G_{\{2\}}(s) = \frac{T_{\{2\}}}{H_{\{2\}}}$$

Objective

Analyse the system and **design** a performant and robust controller for **set point tracking**.

Test

Inputs will be defined during the defense as 4 steps over a 15 minute period. Make sure your implementation can handle this.

Important clarification on system selection

Although the TCLab has two heaters and two sensors, for this project you will work using a **SISO approach**.

You must experimentally analyse both subsystems:

$$G_{\{1\}}(s) = \frac{T_{\{1\}}}{H_{\{1\}}}$$

$$G_{\{2\}}(s) = \frac{T_{\{2\}}}{H_{\{2\}}}$$

Based on your analysis, you must **select the subsystem that performs better** and justify your choice.

Your selection must be supported by experimental evidence. For example:

- quality of the response
- noise levels
- speed of the dynamics
- stability and repeatability
- ease of control

All further work including tuning, analysis, and controller design must be done on the selected subsystem only.

The comparison between **G1** and **G2** is part of the evaluation.

Step 1 Baseline controller

Each group must implement a **baseline controller**. This is a strict requirement to be allowed to defend the project.

The baseline consists of a properly tuned **P, PI, PID, or a close variation**. You may use material from the TCLab webpage for this part.

However, simply running existing code is not sufficient. You must clearly explain:

- which controller you selected
- how you tuned it
- why you tuned it that way
- what the observed response tells you about the system

This baseline is only the starting point. A correct and clearly justified baseline corresponds to **1 out of 20**.

Without a valid and explained baseline controller, the project cannot be defended.

Step 2 System analysis and characterisation

Starting from your baseline, you must analyse and characterise the selected system.

This includes:

- identifying the main dynamics of the system
- discussing limitations and challenges

investigating nonlinearities, delays, saturation, or other effects if present
supporting your conclusions with experiments, plots, and where useful simulations

The goal is to demonstrate a clear engineering understanding of the system beyond simple implementation.

Step 3 Controller design and comparison

Based on your analysis, you must design a **second controller of your choice**.

You are free to choose from your arsenal of control theory controllers or go beyond it. We encourage critical thinking and innovation. However, the goal remains a **performant and robust controller**, and even more importantly, a clear and valuable set of **engineering insights** about the system and your design choices.

The focus is on:

- clear design choices
- engineering reasoning
- justification of decisions
- quality of insights

You must compare your proposed controller **side by side** with the tuned baseline controller.

This comparison must be:

- clear
- quantitative
- supported by plots
- supported by performance metrics

You must explain:

- what improves
- what does not
- why the behaviour is different
- what trade offs exist
- what you learned from the comparison

Important

Using material from the TCLab webpage is allowed for the baseline. However, merely replicating existing code without understanding will be graded accordingly.

Your grade depends on your analysis, insights, design decisions, and your ability to defend them.

Code and analysis that are suspiciously similar to those of your colleagues and that you cannot defend on your own will be treated accordingly.

Report and implementation

In the report you must clearly document:

comparison between **G1** and **G2** and justification of the selected subsystem
baseline controller and tuning procedure
system analysis and characterisation
identified limitations and challenges
design and justification of your proposed controller
side by side comparison with the baseline
critical discussion of performance, robustness, and limitations
main engineering insights obtained during the project

You must come to the defense with your hardware, software, and presentation ready and working.

Required plots

During the defense

Create 2 time synchronised plots arranged vertically using **your selected controller**:

1. **Input vs Output**

Input in black

Output of the real plant in red

2. **Control effort**

Control effort in red

Ensure:

both plots share the same time axis
graphs update in real time
signals are properly scaled for comparison
results are saved after execution
the plotting window remains open for discussion

In the report

You must include a **clear comparison between the baseline controller and your proposed controller**.

This comparison must include:

output of the real plant with the proposed controller in red
output of the real plant with the baseline controller in blue
control effort of the proposed controller in red
control effort of the baseline controller in blue

Ensure:

both controllers are tested under comparable conditions
experiments start from approximately the same initial temperature
the same input profile is used for both controllers
plots are clearly aligned and easy to compare

The comparison must be quantitative, supported by performance metrics, and properly discussed. If this comparison is missing or unclear, the evaluation will be affected accordingly.

Deliverable Rules

Final report including design, implementation details, performance analysis, and recommendations, maximum 10 pages.

Code in python, matlab, or similar, properly commented.

Submit everything as a zip file.

Deadline: **25 May at 23:30**

Send to carlos.rodriguez@kuleuven.be including your presentation slot.

Late submissions deduct **one point per minute**.

Defenses will take place on **Tuesday 26**, **Wednesday 27**, and **Thursday 28**.

Select your slot here:

<https://kuleuven->

[my.sharepoint.com/:x:/g/personal/carlos_rodriguez_kuleuven_be/IQCmDd25o35CQq2V6th9WSbKAQw8R5BUlqbMazmsNDB9kXw?e=gnMaQr](https://kuleuven-my.sharepoint.com/:x:/g/personal/carlos_rodriguez_kuleuven_be/IQCmDd25o35CQq2V6th9WSbKAQw8R5BUlqbMazmsNDB9kXw?e=gnMaQr)

The files you submit are final and cannot be modified after submission.

Assessment Criteria

1. Depth and quality of system analysis and understanding.
2. Quality of controller design, justification, creativity and insights.
3. Performance and robustness of the implemented solution.
4. Clarity of report and defense.